

Swiggy Restaurant Analytics Dashboard

Project Objective

The objective of this project is to develop an end-to-end Business Intelligence solution using SQL Server and Power BI to analyse restaurant performance, customer ratings, menu popularity, and city-wise business trends across the Swiggy platform.

The project aims to transform raw restaurant data into meaningful business insights through data cleaning, SQL analysis, KPI development, and interactive dashboard visualisation, enabling data-driven decision-making for restaurant growth and operational improvement.

Core Business Problem

Swiggy hosts thousands of restaurants across multiple cities, offering a wide variety of cuisines and dishes. While restaurants receive significant customer engagement through ratings and reviews, business stakeholders need to understand which restaurants, cities, dishes, and food categories are driving customer satisfaction and business performance.

The primary objective is to analyse restaurant data to identify high-performing restaurants, customer preferences, popular dishes, and regional trends. These insights help support strategic business decisions, improve customer experience, and enable restaurant partners to optimise their offerings based on data-driven recommendations.

Business Questions

1. Executive Business Performance

Business Question:

How is the overall restaurant ecosystem performing across the Swiggy platform?

Analysis Focus:

- Total Restaurants
- Total Cities
- Total States
- Total Dishes
- Average Rating
- Total Customer Ratings

2. Restaurant Performance Analysis

Business Question:

Which restaurants consistently deliver the highest overall performance based on customer ratings and engagement?

Analysis Focus:

- Top-performing restaurants
- Restaurant Performance Score
- Customer Rating Count
- Average Ratings

3. City Performance Analysis

Business Question:

Which cities contribute the highest restaurant performance and customer engagement?

Analysis Focus:

- City Performance Score
- Restaurant Distribution
- Customer Ratings by City
- Average Ratings by City

4. Category Performance Analysis

Business Question:

Which food categories generate the highest customer engagement?

Analysis Focus:

- Category Rating Count
- Category Popularity
- Category Distribution

5. Dish Popularity Analysis

Business Question:

Which dishes are the most popular among customers?

Analysis Focus:

- Top Rated Dishes

- Dish Rating Count
- Customer Preferences

6. Customer Rating Analysis

Business Question:

How do customer ratings vary across restaurants, cities, and categories?

Analysis Focus:

- Average Ratings
- Rating Distribution
- Customer Engagement Trends

7. Strategic Business Recommendations

Business Question:

What business actions can improve restaurant visibility, customer engagement, and platform performance?

Expected Recommendations:

- Promote high-performing restaurants
- Improve visibility of popular dishes
- Expand successful food categories
- Strengthen restaurant presence in high-performing cities
- Improve customer engagement strategies

Technical Scope

SQL Server

- Data Cleaning
- Data Validation
- Exploratory Data Analysis (EDA)
- KPI Calculations
- Restaurant Performance Analysis
- Customer Rating Analysis
- Category & Dish Analysis
- City-wise Business Analysis
- Aggregate Functions

- Window Functions
- Common Table Expressions (CTEs)

Power BI

- Executive Dashboard
- Restaurant Performance Dashboard
- City Performance Dashboard
- Category Analysis Dashboard
- Dish Popularity Dashboard
- Customer Rating Dashboard
- Interactive Filters & Slicers
- KPI Cards
- Dynamic Visualisations

Skills Demonstrated

Technical Skills

- SQL Server
- Data Cleaning & Validation
- Exploratory Data Analysis (EDA)
- Aggregate Functions
- Window Functions
- Common Table Expressions (CTEs)
- Power BI
- Dashboard Development
- KPI Reporting
- Data Visualisation

Business Skills

- Business Intelligence
- Restaurant Performance Analysis
- Customer Analytics
- Category & Product Analysis
- Geographic Analysis
- KPI Development
- Executive Reporting
- Data-Driven Decision Making

Expected Business Impact

This project enables business stakeholders to:

- Identify top-performing restaurants and cities.
- Understand customer preferences and popular dishes.
- Improve restaurant visibility through data-driven insights.
- Support strategic decisions using interactive dashboards.
- Enhance customer engagement and business performance through KPI monitoring.